Note:

The information contained in this document is intended as an additional support to the Intro to Stats Labs. It contains very (very!) simplified interpretation of complex topics, and in no way supersedes or replace the official materials and information provided in the lecture and course materials.

Use with caution

Simple Linear regression: {Remember these steps!}

1. Identify dependent (aka outcome) and independent variable (aka predictor). {*The outcome needs to be a CONTINUOUS variable. The predictor variable(s) can be of any kind.*}
2. Create a scatterplot. {*The dependent variable goes on the Y-AXIS, the independent variable goes on the X-AXIS*}. Interpret it.
3. Check the assumptions of the linear regression! {*Remember that you have the log tool to use as needed*}
4. Run your regression {*make sure you have identified well which variables are continuous and which categorical*}
5. Interpret your regression results:

Key Element	Use/Interpretation
Coefficients *(β_1, β_2...)*	For every one-unit increase in x, y changes by β amount, *holding all other variables constant*. This "holding constant" part is crucial in multiple regression.
p-values	They tell you whether a coefficient is statistically distinguishable from zero (i.e., whether the predictor likely has a real effect vs. noise). Conventionally, $p < 0.05$ is considered significant.
Confidence intervals	A range where the true coefficient likely falls. Narrow = more precise estimate; wide = more uncertainty.
R^2 *(R-squared)*	The proportion of variance in y explained by your model. An R^2 of 0.72 means your predictors explain 72% of the variation in the outcome. Higher is better, but context matters.
Intercept *(β_0)*	The predicted value of y when all predictors are zero. Often not meaningful on its own (e.g., a "zero height" person), but necessary for the model.

{Coefficients only make sense *within the range of your data* (don't extrapolate wildly)}
{In multiple regression, coefficients change meaning depending on what else is in the model}

Example 1: Predict **how much money (£) an undergrad student spends on a night out** in Edinburgh based on **how many hours they stay out**.

Design: Survey 50 students (N=50) {*Not real data!*}

Regression Equation:
$\hat{y} = \beta_0 + \beta_1 x$ (or with multiple predictors: $+ \beta_2 x_2 + ...$)

Applied Regression Equation:
Outcome/How much you spend ($\hat{y}$) = Intercept (£5.00) + Coefficient (£3.50) + Predictor (Hours)

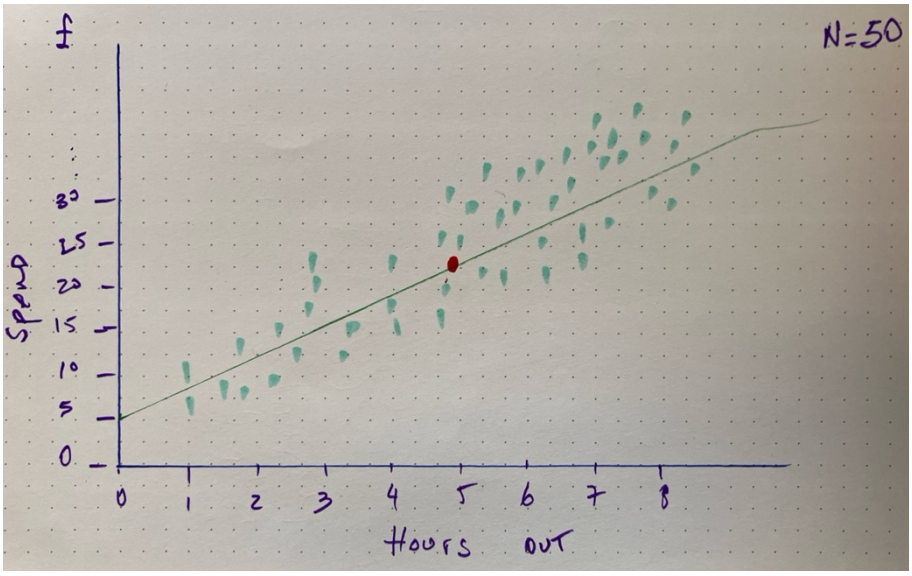
Intercept (£5.00): Even before you've been out a single hour, the model predicts you'll spend £5. Think of this as your entry costs — e-bike rental + first round. On its own it's not always meaningful, but it anchors the line.

Coefficient (£3.50) For every extra hour you stay out, you spend an additional £3.50 on average.

$R^2 = 0.65$ Hours out explains 65% of the variation in spending between students. That's pretty good! The other 35% is down to things not in the model — whether you pre-drink, how generous you are buying rounds, or whether you end up in a Grassmarket cocktail bar vs. a Wetherspoons.

p-value < 0.001 The coefficient on hours is highly statistically significant. It's extremely unlikely this relationship is just random noise in the data.

Briefly, a student who stays out for 5 hours is predicted to spend: £5.00 + (£3.50 × 5) = £22.50



Notes:
Don't extrapolate — the model shouldn't be used to predict spending for a 20-hour night out. That's outside the range of the data (and outside the range of human endurance, especially without food!)
The model is a simplification — one predictor rarely tells the whole story

Multiple Linear regression

Five potential reasons for adding independent variables:

1) **Confounding**

Confounding is about getting an unbiased estimate of your main variable's effect. A confounder is *related to both your predictor and outcome*, so omitting it causes the coefficient on your main variable to be wrong. You add it to correct the estimate, not because you care about the confounder itself.

Example 1: How much does an undergrad student spends on a night out in Edinburgh based on how many hours they stay out?
A confounding variable might be *day of the week* (weekend vs. weekday) affects both how long students stay out *and* how much they spend (more deals, more friends out, bigger rounds, having dinner and cocktails). If you omit it, the effect of hours on spending looks larger than it really is, because it's partly just capturing the weekend/weekday effect. You add it to isolate the true relationship.

2) **Mediation**

Mediation is about unpacking *how a variable has its effect*, it explains the association between the dependent and independent variable. A mediator sits between your predictor and outcome ($X \rightarrow M \rightarrow Y$). You might include or exclude the mediator depending on whether you want the total effect of X or just its direct effect after accounting for the pathway through M.

Example 1: How much does an undergrad student spends on a night out in Edinburgh based on how many hours they stay out? A mediator might be *drinks consumed*. Hours out might increase spending *because* more hours means more drinks consumed. Drinks purchased is the pathway: hours → drinks → spending. If you include drinks in the model, you're asking "does time out *beyond* its effect on drinks still matter?" Exclude it if you want the total effect of hours on spending.

3) **Moderation**

Moderation is about interaction. A moderator is a variable (Z) that affects the direction or the strength between the independent (X) and the dependent variable (Y). You add Z and the product term $X \times Z$ to capture this conditionality. Unlike confounding or mediation, the focus is on how two predictors jointly shape the outcome. Interaction terms are essentially the *mechanics* of how you *implement* moderation in a regression model, you create it by literally multiplying the two variables together (hours(×)group_size) and including that product as an extra predictor alongside the original variables.

Example 1: How much does an undergrad student spends on a night out in Edinburgh based on how many hours they stay out?
The effect of hours on spending might depend on whether the student is going out in a *group or alone*. In a large group, each extra hour out adds a lot to the bill — rounds get expensive fast, people buy each other drinks, and the social pressure to keep spending is high. But a student going out solo might nurse a drink for an hour and barely spend anything extra over time. You add the interaction term hours(×)group_size to capture that the *slope* of hours on spending is steeper for students in larger groups.

* **Flexibility**
Flexibility means allowing a relationship to take a more complex shape than a straight line. Adding polynomial terms (X^2) of the same variable lets the model bend and curve. You're not adding a new substantive variable — you're giving the model more room to fit the functional form of an existing one. {Because it might be that the reason we are not capturing the relationship is that we are assuming a linear shape}

Example 1: How much does an undergrad student spends on a night out in Edinburgh based on how many hours they stay out?
The relationship between hours and spending might not be linear. The first couple of hours might see rapid spending (entry fees, first round), then it plateaus, then spikes again at the end (kebab + cab back home). Adding hours^2 lets the model curve to capture this non-linear pattern rather than forcing a straight line.

* **Prediction**
Prediction is purely pragmatic — you add variables simply because they improve out-of-sample predictive accuracy. You don't care about causal interpretation at all; the coefficients are just levers that make your predictions better. {This is NOT what we aim to do as social scientists!}

Example 1: How much does an undergrad student spends on a night out in Edinburgh based on how many hours they stay out?
You want to predict a student's bar tab at the end of the night. You notice that *which area of Edinburgh* they're in (Grassmarket vs. Newington) improves your forecast, so you add it. You don't particularly care *why* it helps — it just makes predictions more accurate.

The key practical takeaway is that confounding, mediation, and moderation are "causal" concepts that require theoretical justification, while prediction and flexibility are statistical goals that are more agnostic about causation. Mixing up these motivations (specially mistaking a mediator for a confounder!) can lead to badly misinterpreted results.

Example 2: Stress ---> Poor Sleep (More stress leads to worse sleep quality)

1. **Confounder: caffeine intake**
   People who are stressed tend to drink more coffee, but caffeine independently disrupts sleep. So, if I omit caffeine from the model, the effect of stress looks larger than it really is because is partly just picking the effect of the caffeine. By adding caffeine intake to the model I can isolate the true effect of stress on sleep, separate from the coffee habit that tends to come with it

2. **Mediation: rumination** (repetitive negative thinking at bedtime)
   Stress increases rumination, and it's actually the rumination that disrupts sleep. The pathway is stress → rumination → poor sleep
   Rumination carries the effect of stress through to sleep. If I include it in the model I'm estimating the direct effect of stress on sleep above and beyond what goes through rumination. I can exclude it if I want the total effect of stress on sleep

3. **Moderation: Exercise**
   The effect of stress on sleep depends on whether some exercises regularly or not. For people who don't exercise, high stress badly disrupts sleep. For "regular exercisers", the same level of stress has a much weaker effect on sleep. Exercise won't explain how stress affects sleep, but it will change how strong that relationship is. I add the interaction term stress(x)exercise to let the slope differ between the two groups

4. **Flexibility: Stress has a non-linear effect**
   The relationship between stress and sleep might not be a straight line. *Very low* stress might actually help sleep slightly (eg. when you have done a lot and have a mild sense of productive tiredness), moderate stress disrupts it increasingly, but extreme high stress might plateau (the person is so exhausted that they class regardless). I add a $stress^2$ to let the model curve to capture this pattern, rather than forcing a rigid straight line through the data.

5. **Prediction: Bedroom temperature**
   You saw a reel discussing adjusting bedroom temperature to get better sleep. You add it to the model and improves your adjusted R^2. You don't have a particular theoretical reason linking it to stress, it just makes the prediction better. While some people might just care about accuracy and prediction and add it to their models without much thought, as a social scientist I want to understand what is happening under the bonnet of my model so I actually review the theory and figure out what is the best way to include it in my model.